

When the Network and Internet Fails

Internet Privacy

Internet Privacy deals with the fundamental human rights of a person's privacy on the Internet, including the rights to the privacy of an owner's displayed, stored, and confidential information.

The importance of internet privacy is the control over identity and personal information. Without it, anyone with malicious intentions can alter and use someone's identity for their agenda, including stealing money and identity theft.

Different Techniques to Protect Internet Privacy

Without internet privacy, interested third-party companies will collect and analyze every online activity to take advantage of these data. Here are some techniques to protect internet privacy.

Securing Web Browser

Web browsers such as Google Chrome and Mozilla Firefox are the most used browsers online for their specific features and familiarity, although a total security guarantee is not automatic.

Better check reviews and read about their security features before using a web browser. Try using different web browsers for specific periods to compare interfaces and safety features before deciding.

Using a Virtual Private Network (VPN)

It is the best way to have Internet privacy. It does not just change the IP address; it also protects incoming and outgoing Internet traffic with extensive encryption. Online activities and personal information stay secure and private from snoopers because of this technology.

Double-checking Unfamiliar Links

Never trust links from suspicious emails, sketchy websites, and advertisements. Double-check the URL to see if it has the correct format before clicking.

Although simply clicking a link does not mean it will contain malicious and harmful content, it still does impose a more significant risk. Phishing sites often hide behind deceiving links that resemble a trusted brand, which makes double-checking the link an essential habit online.

Limited Sharing on Social Media

Social media platforms like Instagram, Facebook, and Twitter are all publicly accessible. Their privacy protection is optional by locking the account or making it private.

Threats can be avoided by simply limiting too personal posts on social media. Avoid providing information that alludes to key information, such as favorite color or a pet's name, which can be used as clues to passwords and security questions.

Using Multi-factor Authentication

Multi-factor or two-factor authentication is an added verification step besides a conventional password. Google applies this wherein the users must provide a specific code sent to their smartphones to access their accounts.

Applying multi-factor authentication reduces the chances of online sabotage or falling victim to cyber threats and cyber-attack.

Network Security

As network devices exponentially increase, the protocols, standards, and configurations must be defined on how they communicate with each other.

Network Security

It ensures the integrity, confidentiality, and accessibility of computer networks. It is relevant when cybercriminals are becoming more dominant in exploiting new users and enterprises, then it is vital to take extra precautions to avoid any data breach.

Network Technologies

Here are some network security technologies of today:

Firewall

It was previously stated that it monitors and filters incoming and outgoing network traffic. Additionally, it serves as a “*wall*,” either hardware or software, between the trusted internal and untrusted external Internet by only allowing the verified traffic based on the predetermined set of rules in place.

Intrusion Detection System (IDS)

It is a software or a device for closely monitoring a network for any malicious activity or intrusion at the network or the host.

For a Network IDS, a *signature-based detection method* is used wherein it seeks certain patterns of known malicious instruction sequences by targeting new malware sequences. It relies on machine learning to develop a model of trusted activity, then compares it to the current activity.

For the detection of intrusions at the host, a *host intrusion detection system* (HIDS) is used to run on the individual hosts of a network; therefore, it only monitors the inflow and outflow of traffic at the host.

It takes a snapshot of the current file system and compares it to a previously taken snapshot. The administrator is alerted if it finds any modification on the critical files.

Wi-Fi Protected Access 3 (WPA3)

It is the latest wireless network security certificate protocol by the Wi-Fi alliance. It focuses its filtering at the “*entry*” when a new device requests to connect to the network.

It also offers “*forward secrecy*,” that if the server's private key is compromised, the session key will not be affected. So, if the long-term passwords were hacked, the past encrypted sessions cannot be decrypted.

Virtual Private Network (VPN)

It connects any remote device to an enterprise server wherein company data is hosted. It piggybacks on a public network to directly access the server, ensuring that the users' online activities are publicly hidden. This feature makes it one of the most powerful network security tools.

Email Security

As emails allow scripts to be embedded inside the device, it is easier for anyone to write a code, attach it to an email, then send it. Receivers tend to click this email which can comprise the device.

The most effective security methods include email spam filters with deep learning to filter domains that send spam.

Consequences of Internet and Network Failure

Since the internet is a collection of interconnected networks, their failures share the same consequences for an organization. The internet and network fail when privacy and security are not properly monitored or ignored.

Here are the following consequences of combined failures of the internet and network:

Revenue Loss

One of the most significant consequences companies face during an internet, and network downtime is revenue loss. Even if the downtime only lasts minutes, it still puts operations and transactions on hold, which is considered a revenue loss for these companies.

Proper cable management is the best way to ensure network reliability, as well as an occasional internet check-up and monitoring.

Maintenance Cost

Repairing damaged internet and network lines consumes both time and maintenance costs, including acquiring new cables and upgraded network devices for replacement.

Daily monitoring of the cables and network devices is encouraged to prevent sudden company downtime.

Lesser Productivity

Internet and network failure significantly affect productivity as tasks heavily rely on the internet. Employees will remain stagnant until the downtime is fixed, causing low productivity outcomes.

It is advisable to provide a backup network and a backup internet service provider for quick resolutions for sudden downtimes.

Damaged Reputation

If internet and network failures regularly occur, it negatively impacts the employees and the company's clients and partners. Reoccurring internet and network problems could be interpreted as negligence, reflecting bad management and maintenance.

It is encouraged to maintain a system that would ensure the reoccurrence of the problems, and it is advisable to assure everyone of the internet and network security in place.

Legal Repercussions

Failures in operations and transactions mean failure to meet agreed-upon outputs and deliverables on time. It could lead to legal repercussions, especially when crucial outputs are at risk.

The best way to prevent this is by alerting every concerned party as soon as failure happens to lay down plans of recovery without compromising the quality and the demand of the clients.

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